

After [creating a project](#), you will be taken to the **Import data** module.

The spatialGE accepts data from Visium and CosMx-SMI experiments directly exported from Space Ranger and AtoMx, respectively. Users can also import data using the “Generic” format option, which accepts plain text files containing expression and spatial coordinates tables. To read more about the accepted formats, please refer to the [“How to get started”](#) section in the application sidebar.

This tutorial will use a Visium data set containing four samples as an example. The data sets may be downloaded from [here](#). Please note that you may be required to create an account. The data sets are the “Human Breast Cancer (Block A Section 1)”, “Human Breast Cancer (Block A Section 2)”, and “Human Breast Cancer: Ductal Carcinoma In Situ”, and “Human Breast Cancer: Whole Transcriptome Analysis”.

Samples in spatialGE are added one at a time. Hence, the following steps should be repeated until all samples are imported into the project.

In this example project, named “TestProject”, start by assigning a name to the sample. If you choose not to assign a name to the first sample, the name “Sample01” will be automatically assigned. The first sample that will be imported is the ductal carcinoma sample. Hence, the name “ductal\_carcinoma” will be typed in the **Sample name** box.

**Note:** The name of any given sample cannot be contained within the name of another sample. For example, two samples with names “tissue1” and “tissue12” are not allowed. Instead, the user can use “tissue01” and “tissue12”.

Next, the gene expression data will be imported. Since this is a Visium data set, a .h5 file can be dragged from a directory in the computer to the center of the **Gene expression** box. Alternatively, you can click the box to open a window to find and select the .h5 file. The same process should be repeated with the **Coordinates** box, which takes the comma- or tab-delimited file containing the Visium spot coordinates.

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Samples for project: TestProject

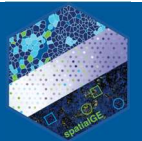
1 Sample name (optional)  
ductal\_carcinoma

Please select or drop your gene expression and coordinates files (tissue image is optional)

| 2                                  | 2                              | 2                        | 2                         |
|------------------------------------|--------------------------------|--------------------------|---------------------------|
| Gene expression<br>+<br>2 required | Coordinates<br>+<br>3 required | Tissue image<br>+        | Scale factor<br>+         |
| Valid file types:<br>h5            | Valid file types:<br>csv       | Valid file types:<br>png | Valid file types:<br>json |

IMPORT SAMPLE

You haven't uploaded any samples yet!



Optionally, you can import a tissue image accompanying the expression and coordinate data. The images can be drag-and-dropped to the **Tissue image** box. In the case of Visium, a JSON file may also be available. Please upload it using the **Scale factor** box. The JSON file is a default output of the Space Ranger algorithm, which is used to appropriately scale the tissue image to the dimensions of the gene expression plots seen in other analysis modules.

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Sample name (optional)  
ductal\_carcinoma

Please select or drop your gene expression and coordinates files (tissue image is optional)

H5  
Visium\_FFPE\_Human\_Breast\_Cancer\_filtered\_feature\_bc\_matrix.h5  
REMOVE

CSV  
tissue\_positions\_list.csv  
REMOVE

Tissue image  
Valid file types: png  
4

Scale factor  
Valid file types: json  
5

IMPORT SAMPLE

You haven't uploaded any samples yet!

To finalize the upload of this sample, click the green button **IMPORT SAMPLE**.

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Sample name (optional)  
ductal\_carcinoma

Please select or drop your gene expression and coordinates files (tissue image is optional)

H5  
Visium\_FFPE\_Human\_Breast\_Cancer\_filtered\_feature\_bc\_matrix.h5  
REMOVE

CSV  
tissue\_positions\_list.csv  
REMOVE

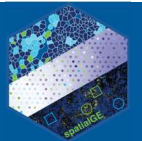
tissue\_hires\_image.png  
REMOVE

JSON  
scalefactors\_json.json  
REMOVE

IMPORT SAMPLE 6

You haven't uploaded any samples yet!

At this point, you will be directed to the project's main page. Three more samples were downloaded from the Visium public data repository. Please do not click the **IMPORT DATA** button yet. Samples cannot be added to a project after clicking the **IMPORT DATA** button. Instead, click the **ADD NEW SAMPLE** button to continue adding samples to the project and repeat the previous steps (1-6 in the figures) for the other two samples.



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**Samples for project: TestProject**

**ADD NEW SAMPLE** ⑦

| Sample            | Expression | Coordinates | Scaling factors | Tissue image |
|-------------------|------------|-------------|-----------------|--------------|
| ductal_carcino... | ✓          | ✓           | ✓               | ✓            |

**Optional: Add sample-level metadata**

- OPTION 1: UPLOAD METADATA FILE (CSV/EXCEL) ?
- OPTION 2: ADD METADATA MANUALLY ?

Sample/Metadata

ductal\_carcinoma

Now that spatial data for all samples has been uploaded, users can input sample-level metadata. This metadata can be any variable(s) associated with the samples or their source. Examples include tissue type, disease, donor, treatment, mutational status, etc. Adding metadata is optional, and if none is available, users can proceed directly to click **IMPORT DATA**.

Two options are available to input sample metadata. Option 1 involves uploading a file containing a table with the metadata. The first column of this file contains the same sample names used to upload the data files (non-matching sample names in the metadata file are ignored). These sample names are case-sensitive. Subsequent columns contain the sample metadata. For more details about this file, please refer to “[How to get started](#)”. Option 2 will be used in this tutorial: Please click the **ADD METADATA MANUALLY** button.

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- Phenotyping

**Samples for project: TestProject**

| Sample              | Expression | Coordinates | Scaling factors | Tissue image |
|---------------------|------------|-------------|-----------------|--------------|
| ductal_carcinoma    | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_vta   | ✓          | ✓           | ✓               | ✓            |

**Optional: Add sample-level metadata**

- OPTION 1: UPLOAD METADATA FILE (CSV/EXCEL) ?
- OPTION 2: ADD METADATA MANUALLY ⑧

Sample/Metadata

ductal\_carcinoma

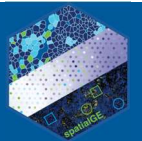
breast\_cancer\_section1

breast\_cancer\_section2

breast\_cancer\_vta

Please add all the samples before clicking IMPORT DATA

**IMPORT DATA**



Next, please click the **ADD NEW METADATA COLUMN** button. An empty column appears in the “Sample/Metadata” table. Now, you can add a name for the first metadata variable and the value of that variable for each sample. The process is repeated for as many times as metadata variables are available.

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| Sample              | Expression | Coordinates | Scaling factors | Tissue image |
|---------------------|------------|-------------|-----------------|--------------|
| ductal_carcinoma    | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_wta   | ✓          | ✓           | ✓               | ✓            |

Optional: Add sample-level metadata

9. ADD NEW METADATA COLUMN METADATA COMPLETE

| Sample/Metadata        | source       |
|------------------------|--------------|
| ductal_carcinoma       | ffpe         |
| breast_cancer_section1 | fresh_frozen |
| breast_cancer_section2 | fresh_frozen |
| breast_cancer_wta      | fresh_frozen |

Please add all the samples before clicking IMPORT DATA

IMPORT DATA

Now that metadata input has been completed, please click the green button **METADATA COMPLETE**.

Sample names can be modified by clicking the pencil icon next to the name. Changes to sample names are not allowed after data import is finished. Please click the **IMPORT DATA** to complete the data import process. A small pop-up will appear asking if you want to receive an email when the data import is finished. The purpose of this email option is to allow users to disconnect and return later for time-consuming processes. This might not be necessary for a small project like this with only three samples.

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| Sample              | Expression | Coordinates | Scaling factors | Tissue image |
|---------------------|------------|-------------|-----------------|--------------|
| ductal_carcinoma    | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_se... | ✓          | ✓           | ✓               | ✓            |
| breast_cancer_wta   | ✓          | ✓           | ✓               | ✓            |

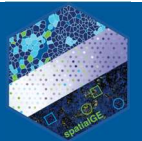
Optional: Add sample-level metadata

ADD NEW METADATA COLUMN 11. METADATA COMPLETE

| Sample/Metadata        | source       | pr_status   | disease |
|------------------------|--------------|-------------|---------|
| ductal_carcinoma       | ffpe         | unknown     | ductal  |
| breast_cancer_section1 | fresh_frozen | pr_positive | ductal  |
| breast_cancer_section2 | fresh_frozen | pr_positive | ductal  |
| breast_cancer_wta      | fresh_frozen | pr_positive | lobular |

Please add all the samples before clicking IMPORT DATA

12. IMPORT DATA



You will be taken to the **QC and data transformation** module when the pop-up disappears. Please click [here](#) to see the documentation for this module.

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QC & Data Transformation

Documentation

- Original summary:** A table displaying the minimum, average, and maximum number of counts per ROI/spot/cell, as well as the minimum, average, and maximum number of genes per ROI/spot/cell. Genes count as expressed if at least one count was detected in at least one ROI/spot/cell.
- Filter data:** Remove ROIs/spots/cells or genes from the data set. The filter can be executed on all samples or a subset. Internally, spatialGE executes the filter in the following order: 1. Samples specified in "Select samples to apply this filter", 2. Genes specified by name in "Filter genes", 3. ROIs/spots/cells and/or genes based on min and max counts.
- Normalize data:** Transform gene expression counts, with log-normal or SCTransform as options. The normalization procedure does not require filtering, but removal of low-count spots/cells or low count genes is recommended.
- Pseudobulk analysis:** Explore sample-level gene expression. Pseudobulk analysis combines gene counts across all spots/cells and produce pseudo-RNAseq samples. This analysis allows exploration of sample-level transcriptional patterns. Does not require execution of the Filter Data or Normalize Data sections.
- Quilt plot:** Generates spatial (quilt) plots showing spot/cell-level metadata. Users can assess QC metrics such as gene counts per spot or total counts per spot and detect potential technical artifacts.

Original summary
Filter data
Normalize data
Pseudo-bulk analysis
Quilt plot

CONTINUOUS SCROLLING

| Sample name            | Total spots/cells | Total genes | Min. counts | Avg. counts | Max. counts | Min. genes | Avg. genes | Max. genes |
|------------------------|-------------------|-------------|-------------|-------------|-------------|------------|------------|------------|
| ductal_carcinoma       | 2518              | 17651       | 253         | 13320       | 43555       | 225        | 4776       | 8414       |
| breast_cancer_section1 | 3813              | 22968       | 524         | 18607       | 72315       | 393        | 5064       | 9594       |
| breast_cancer_section2 | 4015              | 23155       | 96          | 18425       | 80748       | 81         | 5004       | 10020      |
| breast_cancer_wta      | 4325              | 23812       | 35          | 11751       | 59080       | 34         | 3635       | 9059       |